Recap: Algebraic properties of cts functions.
Composition of cts functions.

Ex: The function $f: \mathbb{R} \to \mathbb{R}$ given by $f(x) = \sin(x)$ is cts on $\mathbb{R}$.

Proof: Start by proving that f is cts at $c = 0$.
It's the case that $|\sin(x)| \leq |x| \quad \forall x \in \mathbb{R}$.
↑ Exercise: Unit circle.

It follows that for every $\varepsilon > 0$, letting $\delta = \varepsilon$ we have that

$$|\sin(x)| \leq |x| < \delta = \varepsilon$$

whenever $|x| < \delta$. Let $h: \mathbb{R} \to \mathbb{R}$ be the function $h(x) = \cos(x)$. Using $|\cos(x) - 1| \leq |x|$, $\forall x \in \mathbb{R}$
Unit circle

we have that for every $\varepsilon > 0$, letting $\delta = \varepsilon$ it follows that

$$|\cos(x) - 1| \leq |x| < \delta = \varepsilon$$

whenever $|x| < \delta$. Now the function $g_a : \mathbb{R} \to \mathbb{R}$ given by $g_a(x) = x - a$ is cts at each $c \in \mathbb{R}$ and every $a \in \mathbb{R}$. (Exercise) Since the composition of cts functions is cts, then $f \circ g_a : \mathbb{R} \to \mathbb{R}$ given by $f(g_a(x)) = \sin(x-a)$ is cts at $x = a$ for each $a \in \mathbb{R}$. Since,

$$\sin(x) = \underbrace{\sin(x-a)\cos(a) + \cos(x-a)\sin(a)}_{\text{Double angle formula.}}$$

it follows that $\sin(x)$ is cts for all $x \in \mathbb{R}$. ☐

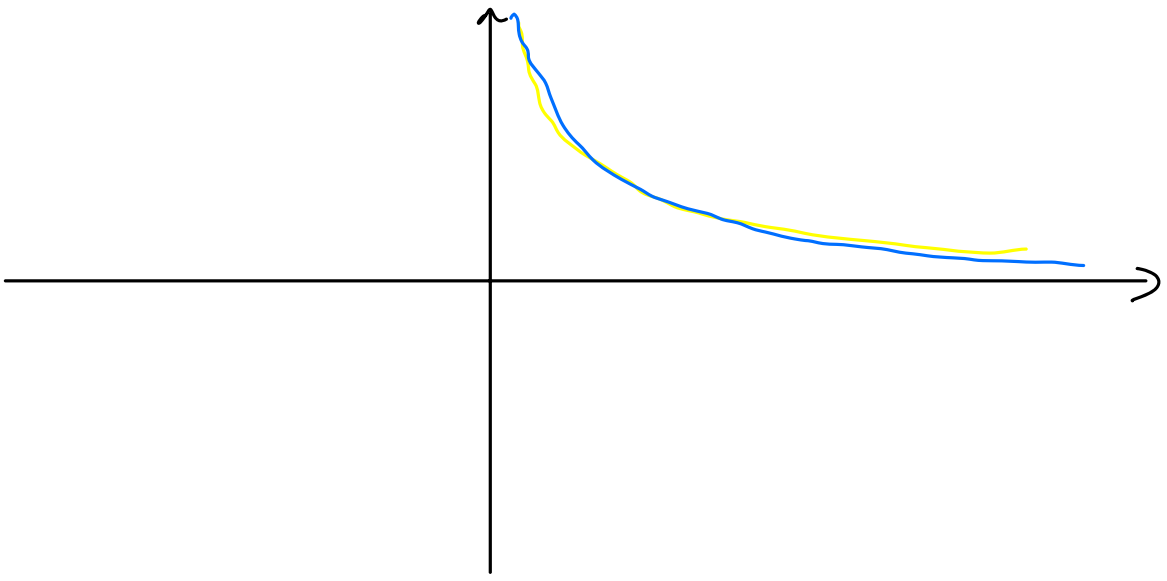
Try proving $\cos(x)$ is cts $\forall x \in \mathbb{R}$.

5 - Discontinuous Functions

Consider the function $f : (0, \infty) \to \mathbb{R}$ given by

$f(x) = \frac{1}{x}$. Now, if our defn of discontinuity was that the function is not cts at a point, this function would be cts everywhere in its domain.

Graph:



If we try to extend this function to say

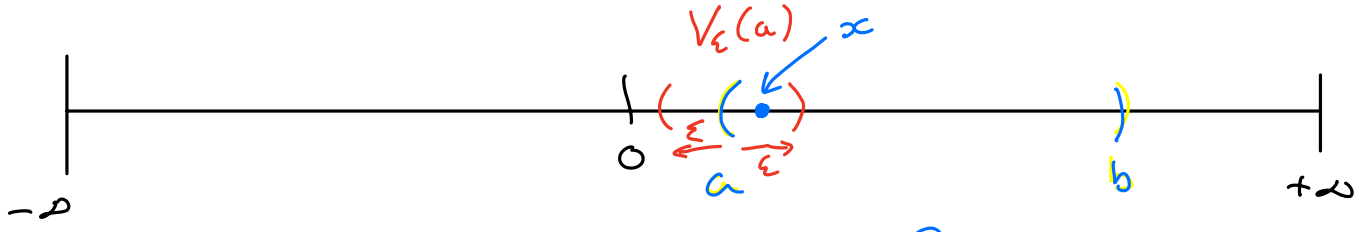
$g: [0, \infty) \longrightarrow \mathbb{R}$ given by

$$g(x) = \begin{cases} \frac{1}{x} : & x \in (0, \infty) \\ p : & x = 0 \end{cases},$$

where $p \in \mathbb{R}$. No matter which value of p we choose, this function is now always not cts at $c = 0$.

Defn: A point $c \in \mathbb{R}$ is a <u>limit point</u> of a set D if for every ε-neighbourhood $V_\varepsilon(c)$ of c there exists a point $x \in V_\varepsilon(c) \cap D$ with $x \neq c$. The <u>closure</u> of the set D, is the union of D and all its limit points. We denote by $\overline{D}$.

Pic:



$D = (a,b) \subset \mathbb{R}$, $\overline{D} = [a,b]$

Defn: A function $f : D \to \mathbb{R}$ is said to be <u>discontinuous</u> at $c \in \overline{D}$ if it is either not cts at $c \in \overline{D}$ or $c \notin D$.

Ex: Let $f : \mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}$ given by $f(x) = 1/x$.

Here, $0 \in \overline{\mathbb{R} \setminus \{0\}}$ and since $0 \notin \mathbb{R} \setminus \{0\}$ the function f is said to be discontinuous at $c = 0$.

Ex: Let $f:[0,1] \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \sin(1/x) : x \in (0,1] \\ 0 : x = 0 \end{cases}.$$

Then f is discontinuous at $c = 0$.

Proof: $(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in D)(|x-c| < \delta \Rightarrow |f(x)-f(c)| < \varepsilon)$

Negating this gives the defn of a function to be not cts at the point c. (Exercise → Propositional Logic)

Let $\varepsilon = 1/2$. Then for ever $\delta > 0$, there exists $n \in \mathbb{N}$ such that

$$0 < \frac{2}{(2n-1)\pi} < \delta.$$

Since,

$$\left| \sin\left(\frac{1}{\frac{2}{(2n-1)\pi}} \right) \right| = \left| \sin\left(\frac{\pi(2n-1)}{2} \right) \right| = 1 \geq 1/2.$$

Hence, f is discontinuous at $c = 0$. ☐

Functional Limits

Defn: (Infinite Limits) Let $f: (a, \infty) \to \mathbb{R}$.

Then *f converges to $L \in \mathbb{R}$ as x goes to infinity* if for every $\varepsilon > 0$, there exists an $M > a$ such that

$$|f(x) - L| < \varepsilon$$

for every $x > M$. We write this as $\lim_{x \to \infty} (f(x)) = L$.

Similarly, let $g: (-\infty, b) \to \mathbb{R}$. Then *$g$ converges to $L \in \mathbb{R}$ as x goes to negative infinity* if $\forall \varepsilon > 0$, $\exists M < b$ such that

$$|g(x) - L| < \varepsilon$$

for every $x < M$. We write this as $\lim_{x \to -\infty} (f(x)) = L$.

Defn: (Limits of functions)

Let $f: D \to \mathbb{R}$ and let $c \in \overline{D}$. Then f converges to $L \in \mathbb{R}$ as x approaches $c \in \overline{D}$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - L| < \varepsilon$$

whenever $0 < |x - c| < \delta$ and $x \in D$. We write this as $\lim_{x \to c} (f(x)) = L$.

Ex: Let $f: \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $f(x) = x \sin(1/x)$.

Then $\lim_{x \to 0} (f(x)) = 0$.

Scratch Work:

$$|f(x) - L| = |x \sin(1/x) - 0|$$

$$= |x| |\sin(1/x)|$$

$$\leq |x| < \delta = \varepsilon.$$

Proof: Given $\varepsilon > 0$, let $\delta = \varepsilon$. Then if $|x| < \delta$ and $x \in \mathbb{R} \setminus \{0\}$ it follows that

$$\begin{aligned} |f(x) - 0| &= |x \sin(1/x)| \\ &\leq |x| \\ &< \delta \\ &= \varepsilon. \end{aligned}$$

□

Theorem 6.4: (Limits and Continuity)

Let $f: D \to \mathbb{R}$. The function is cts at $c \in D$ iff $\lim_{x \to c} (f(x)) = f(c)$.

Proof: Exercise.

Theorem 6.5: (Algebraic Properties of Limits)

Let $f: D \to \mathbb{R}$ and $g: D \to \mathbb{R}$ be real functions such that $\lim_{x \to c} (f(x)) = L$ and $\lim_{x \to c} (g(x)) = M$ for

Some $c \in \overline{D}$. Then,

i) $\lim_{x \to c} (\lambda f(x)) = \lambda L$, $\forall \lambda \in \mathbb{R}$.

ii) $\lim_{x \to c} (f(x) + g(x)) = L + M$.

iii) $\lim_{x \to c} (f(x) g(x)) = L \cdot M$

iv) $\lim_{x \to c} \left(\frac{f(x)}{g(x)} \right) = L/M$ providing $g(x) \neq 0$ $\forall x \in D$ and $M \neq 0$.

Proof: Exercise.

Defn: (One-sided Limits)

Let $f: D \to \mathbb{R}$ and $c \in \overline{D}$. Then f converges to $L \in \mathbb{R}$ as x approaches $c \in \overline{D}$ from the right if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - L| < \varepsilon$$

Whenever $c < x < c + \delta$ and $x \in D$.

We write this as $\lim_{x \to c^+} (f(x)) = L$.

Similarly for $\lim_{x \to c^-} (f(x)) = L$, just changes to

$c - \delta < x < c$.

Ex: Let $f: \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} \sqrt{x} : x \geq 0 \\ -x - 1 : x < 0 \end{cases}.$$

Then $\lim_{x \to 0^+} (f(x)) = 0$ and $\lim_{x \to 0^-} (f(x)) = -1$.

Scratch Work: When $x \geq 0$ we have

$$|f(x) - 0| = |\sqrt{x} - 0|$$

$$= |\sqrt{x}|$$

$$= \sqrt{|x|} < \sqrt{\delta} \quad (\delta = \varepsilon^2).$$

If $f(x) < 0$, we have

$$|f(x) + 1| = |-x - 1 + 1|$$

$$= |-x|$$

$$= |x| < \delta = \varepsilon .$$

Proof: Given $\varepsilon > 0$, let $\delta_1 = \varepsilon^2$ and $\delta_2 = \varepsilon$. Then if $0 < x < \delta_1$, it follows that

$$|f(x) - 0| = \sqrt{|x|}$$

$$< \sqrt{\delta}$$

$$= \sqrt{\varepsilon^2}$$

$$= \varepsilon .$$

Similarly, $-\delta_2 < x < 0$ we have that

$$|f(x) - 0| = |x|$$

$$< \delta_2$$

$$= \varepsilon .$$

Hence, $\lim_{x \to 0^+} (f(x)) = 0 \neq -1 = \lim_{x \to 0^-} (f(x))$.

Thus $\lim_{x \to 0} (f(x))$ does not exist and hence

$f(x)$ is not cts at $c = 0$. ☐.